

Analog CMOS IC Design (EE651)

Assignment- 3

Total Marks – 10 points

Date: 29/08/2025

Submission Date: 5/09/2025 (11:55 PM)

Note: Below is the list of questions on **MOS Transistors** and **Small-Signal Parameters of a MOS Transistor**. You are required to:

- Perform the simulations using **Cadence Virtuoso** with **SCL 180 nm PDK**.
- Prepare a detailed report based on your work.
- Submit the report for evaluation.

Report Guidelines:

- Include a **Conclusion** section at the end, briefly summarizing your insights and observations from the design process.
 - Name the report file as: RollNumber_Assignment3.pdf
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1. Design Common Source (**CS**) amplifiers with active load for the following specifications.
Voltage Gain > 100 V/V, **Power dissipation** < 10 μ W, min input voltage swing = 100 μ V, max input voltage swing = 12mV.
 - a. Calculate the output swing voltage and gain for the minimum and maximum input swings using transient analysis, and also plot the transient response.
 - b. Plot frequency response (both Gain and Phase response) of the amplifier. Extract Bandwidth and Unity Gain Frequency (UGF) and explain the results.
2. Design a Common Drain (**CD**) amplifier with active load for the following specification.
Voltage gain > 0.95 V/V, **Power dissipation** < 2 mW, **Output impedance (Z_{out})** < 100 Ω .
min input voltage swing = 20 mV, max input voltage swing = 1 V.
 - a. Calculate the output swing voltage and gain for the minimum and maximum input swings using transient analysis, and also plot the transient response.
 - b. Plot frequency response (both Gain and Phase response) of the amplifier. Extract Bandwidth and Unity Gain Frequency (UGF) and explain the results.

Note: You have only one available **power supply (VDD of 1.8V)** and a reference **current source of 1 μ A**. The biasing needs to be done with the help of current mirrors and voltage divider circuits using MOS transistors (as in Assignment 2).

Report should include:

- Transistor level circuit diagram (full system and individual sub-blocks) and sizing (with overdrive voltage) table of all transistors.
- Step-by-step design methodology/derivations/calculations (in brief).
- Overall power consumption and its block-wise breakup should be provided.
- Plots should be neatly labelled, and labels should be clearly visible
- Final results should be provided in tabular format
- Any other information as deemed necessary